Channel assignment and voice QoS in synthetic 802.11 networks: model, methods and verification

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Abstract

This note describes a Julia model (`WiFiAssignment.jl`) that assigns channels to 802.11 access points so as to maximise proportional fairness of client throughput, optionally together with the quality of voice calls carried over the same network. It covers the radio and airtime model, the subset-table formulation that makes exact optimisation possible for up to 18 access points, the heuristics, the voice queue and MAC model, the voice-aware objective, and how each component is checked against an independent reference. Results are in the companion `results.pdf`.

1 Network model

Geometry and propagation. A world has $n \le 18$ access points (APs) placed uniformly in an $A \times A$ square, each serving m clients placed at a uniform distance of 5–15 m and uniform angle from it (clipped to the square). Received power over distance d is

$$P\ [\mathrm{dBm}] = p_0 - \alpha \log_{10} \max(d, 1) + \sigma Z, \qquad Z \sim \mathcal{N}(0,1),$$

drawn independently per AP–client link and once per AP pair (symmetric). Defaults are $p_0 = -30$ dBm, $\alpha = 35$ (path-loss exponent 3.5), $\sigma = 4$ dB, noise $N_0 = -95$ dBm, bandwidth $B = 20$ MHz. Two APs *sense* each other when their mutual received power is at least the carrier-sense threshold (-82 dBm); co-channel APs that do not sense each other are *hidden* from one another.

Client rates. An assignment x gives each AP one of k orthogonal channels. For client c of AP j, with signal S_c, let $\mathcal{S}_j(x)$ and $\mathcal{H}_j(x)$ be the co-channel APs that j senses and does not sense, and $a_i \in [0,1]$ the airtime of AP i. Hidden APs interfere in proportion to their airtime; sensed APs share airtime:

$$\phi_c = \min\Big(R_{\max},\ B \log_2\Big(1 + \frac{S_c}{N_0 + \sum_{h \in \mathcal{H}_j(x)} a_h P_{c,h}}\Big)\Big),$$

$$r_c = \min\Big(D,\ \max\Big(\frac{\eta\, \phi_c}{m\,\big(1 + \sum_{s \in \mathcal{S}_j(x)} a_s\big)},\ r_{\min}\Big)\Big),$$

with PHY cap $R_{\max} = 144$ Mbps, MAC efficiency $\eta = 0.72$, per-client demand $D = 40$ Mbps and floor $r_{\min} = 0.5$ Mbps. The objective is proportional fairness [2],

$$U(x) = \sum_c \log r_c \quad \text{(nats)}.$$

Airtime as a fixed point. An AP's airtime is what its clients' demand needs at the rates they actually get:

$$a_j = \min\Big(1,\ \sum_{c \in j} \frac{D}{\eta \max(\phi_c(a), r_{\min})}\Big).$$

Because ϕ_c falls as hidden APs' airtime rises, the right-hand side is monotone in a, so iterating from the interference-free airtime converges to the least fixed point. When every AP already needs all its airtime (the default demand with three or more clients per AP), the fixed point is $a \equiv 1$ and the iteration is skipped. Like every iterative solver in the code, it raises an error rather than return an unconverged value.

2 Optimisation

Separability. Every term in r_c depends only on the set G of APs sharing AP j's channel, so $U(x) = \sum_{\text{channels}} f(G_{\text{ch}})$ with $f(G) = \sum_{j \in G} \sum_{c \in j} \log r_c(G)$. The 2^n values of f are tabulated once; any assignment then costs k lookups and moving one AP costs four.

Building the table. For saturated worlds interference and blocked airtime add over APs, so each client needs subset-sum tables over the low and high halves of the AP index (and each AP one table of blocked airtime); entries are formed row by row in parallel, with up to 16 client rates multiplied before one logarithm. For unsaturated worlds each entry solves its own airtime fixed point. Both paths are deterministic and give bit-identical results for any thread count.

Heuristics.

- *Graph colouring* (baseline): conflict graph at -75 dBm, APs in decreasing degree, least-conflicting colour.
- *Greedy*: APs in decreasing order of the power they put on other APs' clients, each placed on the channel with the largest marginal table gain; optionally followed by *local search* over single-AP moves and pairwise swaps until neither improves the objective.
- *Simulated annealing*: random start, random single-AP moves with Metropolis acceptance and geometric cooling to $T_0/100$. T_0 is calibrated so the mean downhill move from the start is accepted with probability $1/2$ (from 200 sampled moves), and the best assignment is finished with local search.
- `optimize_assignment`: iterated local search. Eight restarts (the greedy order, then random orders), each followed by 200 kicks that reassign 2–4 random APs and re-run local search, keeping the result when it is no worse.

Exact search. Let $B_j(M)$ be the best split of AP set M into at most j channels. The channel holding the lowest AP of M is some $G \subseteq M$, so

$$B_1(M) = f(M), \qquad B_j(M) = \max_{G \ni \min M,\, G \subseteq M} \big[f(G) + B_{j-1}(M \setminus G) \big],$$

and the optimum is $B_k(\{1, \dots, n\})$. AP 1 always opens the first channel, so the intermediate layers only need sets without it, and the top layer only the full set: about $(k-2)\,3^{n-1}/2$ operations, parallel over sets. When $k \geq n$ every AP gets its own channel, which is optimal because adding an AP to a channel can only lower its members' rates.

3 Voice model

Traffic and queues. Each AP carries K two-way calls, i.e. $2K$ flows that each send one packet per 20 ms frame of $b \cdot 20\,\text{ms} + 480$ bits at codec rate b (64 kbps nominal, 12 kbps minimum), with a 150 ms deadline. An AP's raw capacity per frame is the sum of its clients' rates. Background traffic offers a fraction ρ of that capacity. Under FIFO, voice and background of a frame interleave, so each frame enters the queue as one mixed item and voice loses in proportion to its share when the queue overflows; under priority scheduling only voice is queued. Work older than the deadline is dropped from the head of the queue, and delay is measured at the time the packet's bits finish within the frame.

MAC. AP j's utilisation is $u_j = \min(1, \rho + \text{voice bits/raw capacity})$. A transmission collides with probability

$$p_j = 1 - \big(1 - p^{\text{B}}(1 + \textstyle\sum_{s \in \mathcal{S}_j} u_s)\big) \prod_{h \in \mathcal{H}_j} \big(1 - \sigma_{jh}\, q(u_h)\big),$$

where $p^{\text{B}}(n)$ is Bianchi's DCF fixed point [1] for n contenders, $p = 1-(1-\tau)^{n-1}$, $\tau = 2/(1+W+pW\sum_{i<m}(2p)^i)$ with $W = 16$ and $m = 6$ backoff stages (802.11 OFDM), solved by a bracketed Newton iteration. A hidden AP busy a fraction u of the time overlaps a frame with probability $q(u) = u + (1-u)\big(1 - e^{-u/(1-u)}\big)$ (busy at its start, or starting during it, with frames of equal length and exponential idle gaps), and σ_{jh} is the share of j's clients whose signal is within the 10 dB capture threshold of h's. With $R = 2$ retries, service is capacity divided by $\sum_{i=0}^{R} p_j^i$ expected attempts, and a fraction p_j^{R+1} of served packets is lost.

Policies. `:fifo` and `:deadline` use the nominal codec; `:adaptive` (FIFO) and `:combined` (priority) move the codec 35 % of the way per frame towards the rate at which voice fits the airtime left to it, the service minus background under FIFO and the whole service under priority, so the codec only shrinks when voice does not fit.

Voice-aware objective. An AP's voice outcome depends only on the APs sharing its channel, so it adds a second term to $f(G)$. For each G the steady state of the simulator (utilisation, collisions and codec rate) is solved as a joint fixed point, and each flow contributes $\log(b/b_{\text{nominal}} \cdot \text{delivered fraction})$, so codec quality and loss both count and data clients and voice flows are treated alike by proportional fairness. The same heuristics and exact search then apply unchanged.

4 Verification

Each component is checked against a reference that shares no code with it. The counts below come from the deterministic checks in `benchmarks/stress.jl` and `benchmarks/dense.jl`; the other figures are produced by `paper/results.jl`.

- rates, scores and the airtime fixed point against an independent dB-domain implementation with its own Jacobi iteration: exhaustive search over 1–5 channels and up to 6 APs (5 850 checks), every table entry of 40 worlds (8 304), and 96 worlds of 1–18 APs (14 112);
- exact search and `optimize_assignment` against an independent subset dynamic program for 2–5 channels (8 376 checks for 3 channels, 9 936 for 2, 4 and 5) and against each other at 18 APs for 3, 6, 12 and 18 channels;
- the voice queue against the closed-form steady state of an isolated AP (48 cases, loss within 0.17 points), and packet accounting on 40 random worlds;
- Bianchi's model against a slot-level DCF simulation (within 0.005 up to 4 contenders, the most the voice scenarios reach) and the hidden overlap formula against simulation (within 0.002);
- the steady-state voice estimate against the simulator (loss within 0.49 points, codec within 0.10 kbps), and the voice-aware optimizer against exhaustive search (80 cases).

5 Changes made to the original code

- *Subset table*: rebuilt around per-client half-mask subset sums, one logarithm per group of client rates and parallel rows; it reproduces the original table to within 4×10^{-13}.
- *Heuristics*: local search after greedy, calibrated temperature and final polish for annealing, and the new iterated local search `optimize_assignment`.
- *Exact search*: from three channels and 14 APs to any number of channels and 18 APs.
- *Airtime*: from the interference-free demand to the fixed point above.
- *Voice*: the adaptive codec now accounts for background traffic and only shrinks when voice does not fit; background work expires instead of accumulating; FIFO mixes voice and background in proportion instead of placing voice in the middle of each frame; delay is measured within the frame instead of at frame boundaries; the collision model $1 - e^{-0.12\,|\mathcal{H}_j|\,u_j}$ (capped at 0.65) is replaced by Bianchi contention plus capture-weighted hidden-terminal overlap.
- *Voice-aware assignment* and loud failure of every iterative solver.

6 Limitations

Worlds, traffic and shadowing are synthetic and static, and channels are orthogonal. The voice model is an aggregate work queue rather than a packet-level 802.11 simulation. Bianchi assumes saturated contenders, so partly busy APs enter as fractional contenders, and each AP's own clients count as a single contender, which likely understates collisions in call-heavy cells. The exact methods are limited to 18 APs by the 2^n table. Conclusions hold for this model; external validation needs ns-3, packet traces or hardware.

References

[1] G. Bianchi, "Performance analysis of the IEEE 802.11 distributed coordination function," *IEEE Journal on Selected Areas in Communications*, vol. 18, no. 3, pp. 535–547, 2000.

[2] F. Kelly, "Charging and rate control for elastic traffic," *European Transactions on Telecommunications*, vol. 8, no. 1, pp. 33–37, 1997.